Hermitian matrices: $A \in M_n(\mathbb{C})$

A Hermitian if $\boxed{(\bar{A})^t \equiv A^* \equiv A^\dagger} = A$

different notation

(anti-Hermitian if $A^* = -A$)

A^* is called

- Hermitian conjugate matrix
- adjoint matrix

Hermitian matrices are also called self-adjoint

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related to inner products: if $\langle \cdot, \cdot \rangle$ is the inner product on $\mathbb{C}^n$

$\langle x, y \rangle = x^* y$

$\langle Ax, y \rangle = \langle x, A^* y \rangle$ (always true)

$\parallel \qquad\qquad \parallel$

$(Ax)^* y = x^* A^* y = x^* (A^* y)$

could define A^* as the matrix such that

$\langle Ax, y \rangle = \langle x, A^* y \rangle \quad \forall x, y \in \mathbb{C}^n$

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ex $\quad n = 2 \qquad A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad x = \begin{pmatrix} a \\ b \end{pmatrix} \qquad y = \begin{pmatrix} c \\ d \end{pmatrix}$

define A^* by $\langle Ax, y \rangle = \langle x, A^* y \rangle \quad \forall x, y$

$$Ax = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix} \qquad \langle Ax, y \rangle = \begin{pmatrix} b \\ 0 \end{pmatrix}^* \begin{pmatrix} c \\ d \end{pmatrix} = (\bar{b} \;\; 0) \begin{pmatrix} c \\ d \end{pmatrix} = \bar{b} c$$

$$A^* = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \qquad A^* y = \begin{pmatrix} \alpha c + \beta d \\ \gamma c + \delta d \end{pmatrix}$$

$$\langle x, A^* y \rangle = (\bar{a} \;\; \bar{b}) \begin{pmatrix} \alpha c + \beta d \\ \gamma c + \delta d \end{pmatrix} = \bar{a} \underbrace{(\alpha c + \beta d)}_{=0} + \bar{b} \underbrace{(\gamma c + \delta d)}_{=1}$$

$\Rightarrow A^* = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} = \bar{A}^t \neq A$

<u>unitary</u> $A \in M_n(\mathbb{C})$ is unitary if $A^{-1} = A^*$

inner product: A unitary $\Leftrightarrow$ $\boxed{\langle Ax, Ay \rangle = \langle x, y \rangle \ \forall x, y}$

<u>in QM</u> physical states are represented by vectors in a complex Hilbert space

- states are not physical (not measurable)
- the physical quantities (observables) are the expectation values of self-adjoint operators

expected

suppose system is in state $|\psi\rangle \in \mathcal{H}$

if a is a physical quantity, we associate to it an operator on $\mathcal{H}$

$a \to A \in \mathrm{Lin}(\mathcal{H}, \mathcal{H})$

expectation value: $\langle A \rangle \equiv \dfrac{\langle \psi | A | \psi \rangle}{\langle \psi | \psi \rangle}$ $\in \mathbb{R}$ $\longrightarrow \langle \psi, A\psi \rangle$

identify $\langle A \rangle$ with the average measured value of a in state $|\psi\rangle$

why self-adjoint? $\Rightarrow$ so that $\langle A \rangle \in \mathbb{R}$

$\overline{\langle x, y \rangle} = \langle y, x \rangle$ property of inner products

$\langle \psi, A\psi \rangle \in \mathbb{R} \ \forall \psi \Leftrightarrow \overline{\langle \psi, A\psi \rangle} = \langle \psi, A\psi \rangle$ $\Leftrightarrow A = A^*$

$\overline{\langle \psi, A\psi \rangle} = \langle A\psi, \psi \rangle = \langle \psi, A^*\psi \rangle$ $\leftarrow \forall \psi$

$|\psi\rangle$ and $\lambda|\psi\rangle$, $\lambda \neq 0$ represent the same physical state

physical space of states: $\mathbb{P}\mathcal{H}$ projective Hilbert space

another number we can get from $\mathcal{H}$:

<u>transition amplitude</u> if $|\psi\rangle, |\psi'\rangle \in \mathcal{H}$, probability of measuring state ψ as state ψ' is $\dfrac{|\langle\psi|\psi'\rangle|}{\sqrt{\langle\psi|\psi\rangle\langle\psi|\psi'\rangle}}$

GNS construction

$(xy)^* = y^* x^*$

unitary matrix in QM: use them for transformations of $\mathcal{H} \to \mathcal{H}$ that preserve the structure

<u>in terms of symmetries</u>: $T: \mathcal{H} \to \mathcal{H}$ invertible

$$\frac{|\langle T(\psi), T(\psi')\rangle|}{\|T(\psi)\|\,\|T(\psi')\|} = \frac{|\langle\psi,\psi'\rangle|}{\|\psi\|\,\|\psi'\|} \quad \forall\, \psi, \psi'$$

<u>Wigner's theorem</u>:

- T linear and unitary → continuous (up to a non-zero scaler)
- T antilinear and anti-unitary (up to a non-zero scaler) → discrete

→ T is unique up to multiplication by a non-zero scaler

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if G symmetry group, we associate to $g \in G$ an operator $\rho(g)$ on $\mathcal{H}$

because $\rho(\mathbb{1})$ could be a multiple of identity operator, not technically a rep.

→ projective representation: $\rho: G \to \mathrm{Aut}(\mathcal{H}, \mathcal{H})$

$$\rho(g_1 g_2) = c(g_1, g_2)\,\rho(g_1)\,\rho(g_2) \quad / \quad \rho(g_1 g_2) \propto \rho(g_1)\,\rho(g_2)$$

spin example

$\rho_n : U(1) \to \mathrm{Aut}(\mathbb{C})$

rep of $U(1)$: $\rho_n(e^{i\alpha})|\psi\rangle = e^{in\alpha}|\psi\rangle$ $\quad |\psi\rangle \in \mathbb{C}$

$\rho_n(e^{i2\pi}) = \mathbb{1}$ ok if $n \in \mathbb{Z}$ $\quad \| \quad \rho(e^{i\alpha}e^{i\beta}) = \rho(e^{i\alpha})\rho(e^{i\beta})$

$\to$ only $n \in \mathbb{Z}$

what if $n = 1/2$? $\quad \rho_{1/2}(\overset{=1}{e^{i2\pi}})|\psi\rangle = e^{i\pi}|\psi\rangle = -|\psi\rangle$

$\rho_{1/2}(e^{i2\pi}) = -\mathbb{1}$ not a rep. of $U(1)$

if we are doing Q.M., $\rho_{1/2}$ is still acceptable!

$\rho_{1/2}(e^{i2\pi}) \propto \mathbb{1}$ which is enough

$[\rho_{1/2}(e^{i2\pi})|\psi\rangle] \equiv [|\psi\rangle]$

in fact for $U(1)$ ρ_n is a proj rep. $\forall n \in \mathbb{R}$

$\pi : \mathcal{H}\setminus\{0\} \to P\mathcal{H} = \mathcal{H}\setminus\{0\}/\sim \qquad \psi \sim \psi'$ if $\psi' = \lambda\psi$, $\lambda \neq 0$

$\psi \mapsto [\psi]$

$\mathrm{Aut}(\mathcal{H}) \to \mathrm{Aut}(\mathcal{H})/\sim$

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action on quotient space: $\rho(g)[|\psi\rangle] := [\rho(g)|\psi\rangle]$

$$\rho(g_1 g_2)[|\psi\rangle] = [\rho(g_1 g_2)|\psi\rangle]$$
$$= [c(g_1, g_2)\rho(g_1)\rho(g_2)|\psi\rangle]$$
$$= [\rho(g_1)\rho(g_2)|\psi\rangle]$$
$$= \rho(g_1)\rho(g_2)[|\psi\rangle]$$

<u>if we're lucky:</u> when we take the exponential of a Lie algebra rep, we get a projective Lie group rep.

$\Rightarrow$ actual rep

$\underline{so}(3) \cong \underline{su}(2)$ $\qquad$ $SO(3) \not\cong SU(2)$

$\downarrow$ not simply connected $\qquad$ $\downarrow$ simply connected

$\underline{u}(1) \cong \mathbb{R}$ (as Lie algebras)

$so(2,1)$ $\rightarrow$ representations split in 3 classes $\begin{cases} \text{light-like} \\ \text{space-like} \\ \text{time-like} \end{cases}$

J_x, J_y, J_z

$\rightarrow$ next time: talk about light-cone structure in Lie algebra